

OASIS SCHOOL SCIENCE FAIR 2030

THE DIAGNOSTICS CREW

AI-Integrated Medical Prediction Platform

MEET THE CREW

Deekshith

Advith

Manesh

Sana Akshaya

Aleekhya

| OUR PLATFORM

Access the Future

Our website provides real-time predictive analysis for diverse medical imaging data, bridging the gap between detection and diagnosis.

LIVE AT:

medical-prediction-94d.caffeine.xyz



| THREE DIAGNOSTIC PILLARS



DERMATOSCOPY

Surface-level skin analytics
and micro-structure
visualization.



MRI

Magnetic Resonance
Imaging for internal organ
profiling.



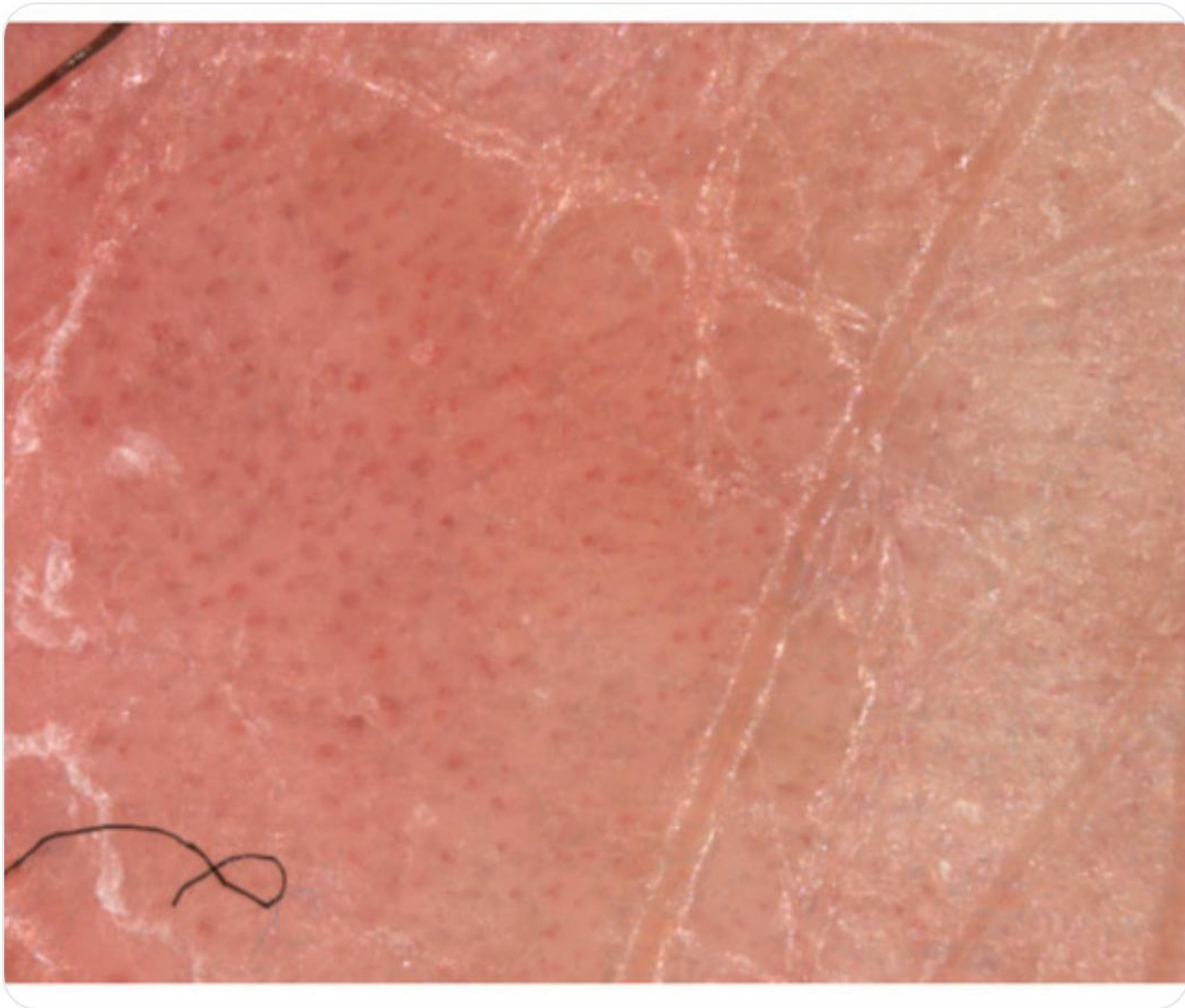
OPG X-RAY

Panoramic dental and jaw
skeletal examination.

SECTION 01

DERMATOSCOPY

VISUALIZED KEY AREAS



Seeing the Unseen

Dermatoscopy highlights features invisible to the naked eye:

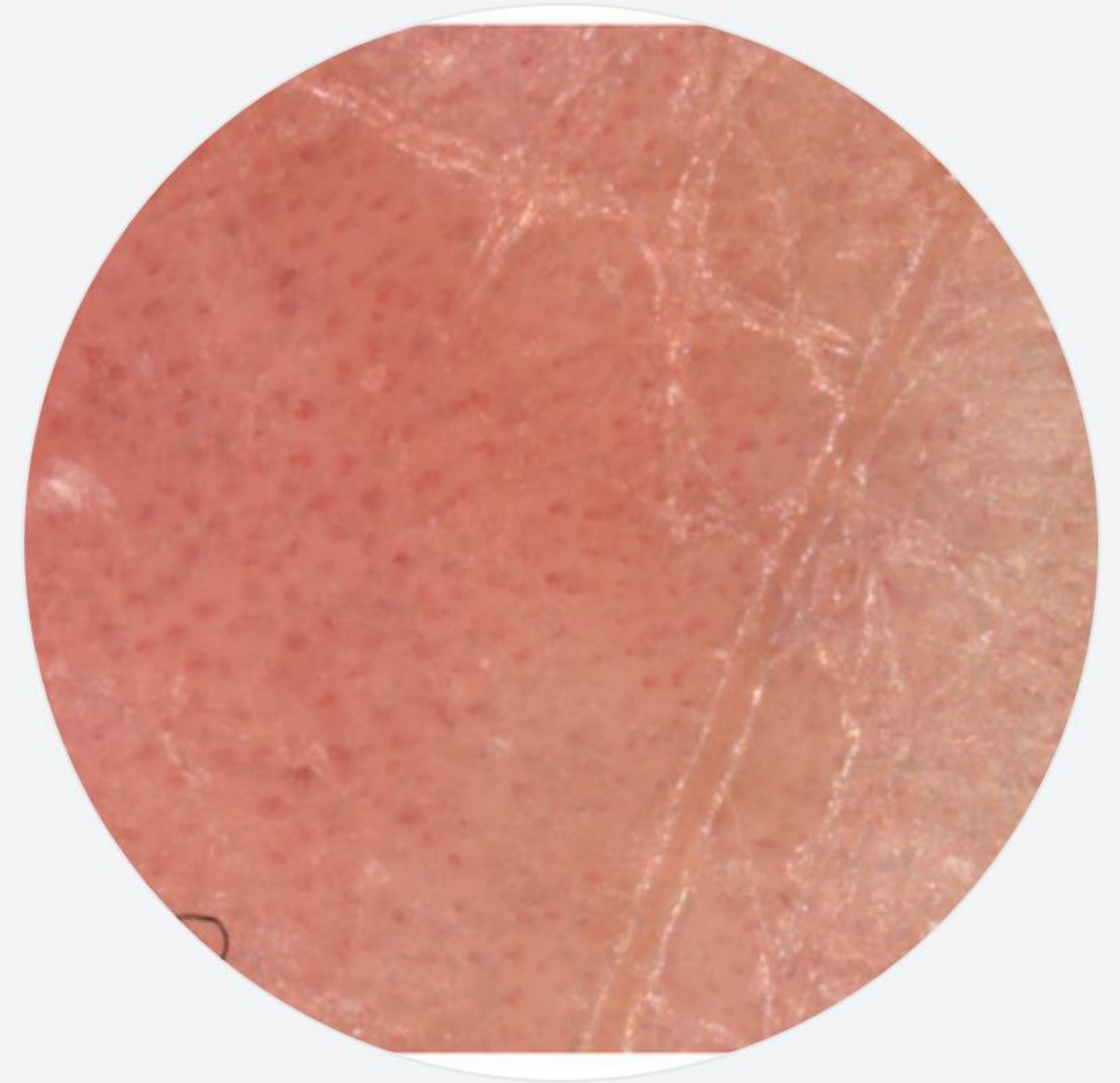
- Skin surface patterns and networks
- Pigmentation variations in depth
- Blood vessels and tiny micro-structures
- Malignant vs Benign lesion markers

| NORMAL SKIN

Healthy Baseline

Normal skin appears even in color and texture. There are no unusual patterns or irregular structures identified by the AI.

- Uniform color distribution
- Smooth and consistent texture
- Absence of abnormal networks

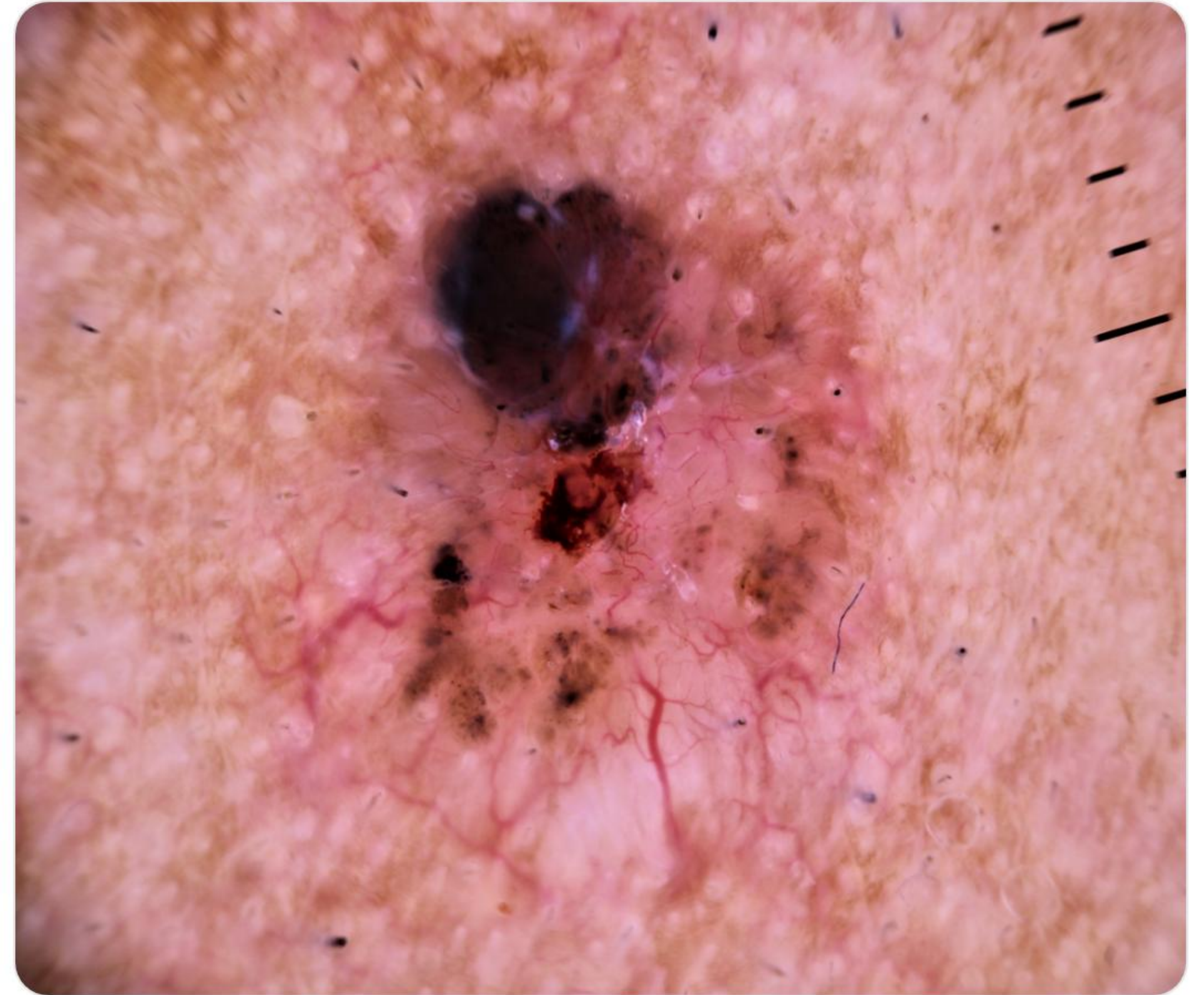


MELANOMA

Serious Skin Cancer

Melanoma appears as an irregular and uneven skin lesion. It requires immediate medical intervention.

- **Asymmetrical:** Uneven shape on both sides
- **Uneven Borders:** Jagged or blurred edges
- **Multiple Colors:** Mixture of brown, black, or red

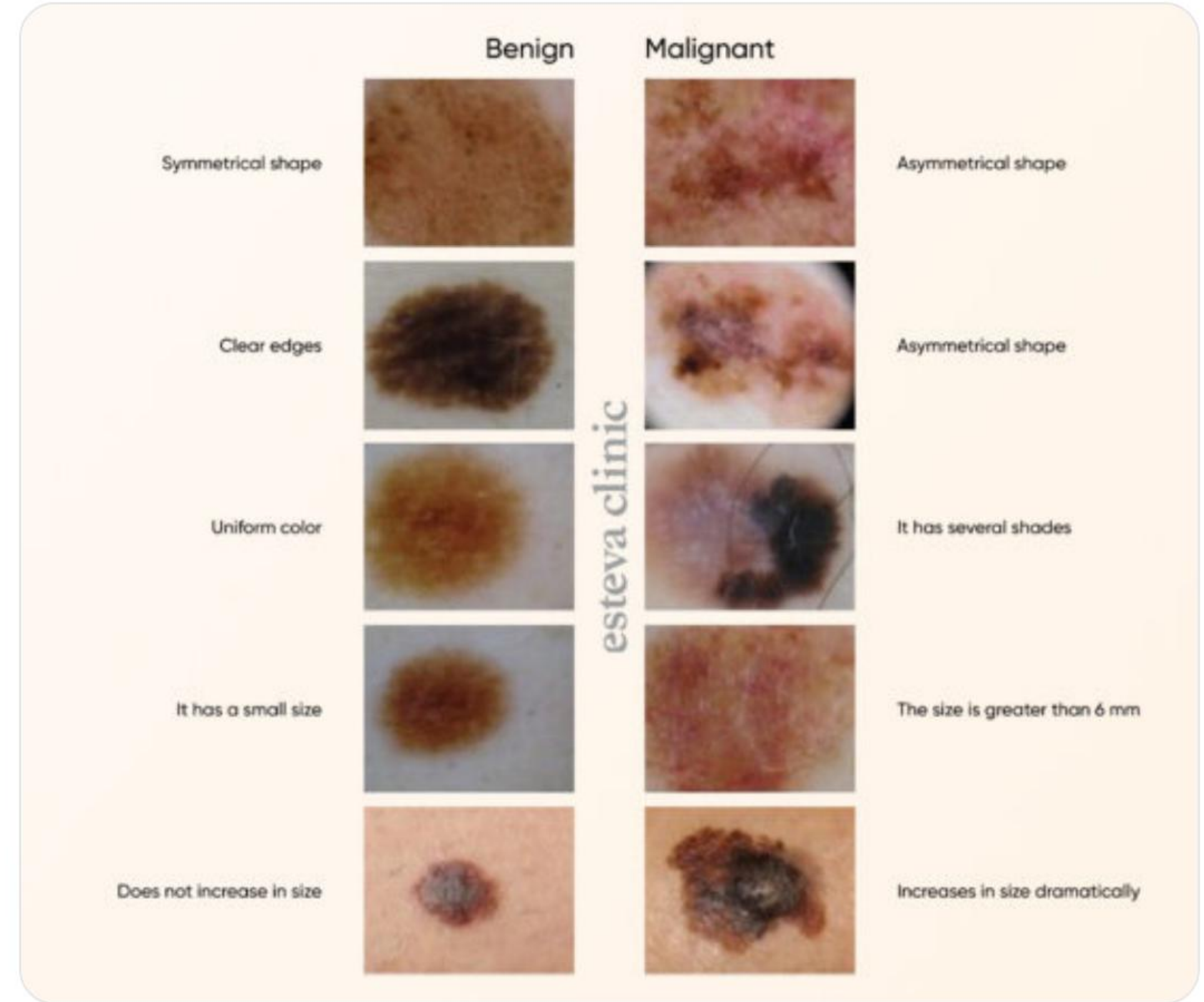


BENIGN MOLE

Non-Cancerous Growth

Benign moles contrast sharply with melanoma in their clinical presentation.

- Symmetrical and circular shape
- Smooth and clearly defined borders
- Consistent tan or brown color



BASAL CELL CARCINOMA (BCC)



Photo provided by Dr. Richard P. Usatine, U

The Pearly Lesion

BCC is common but dangerous if untreated, as it damages surrounding tissue.

- Shiny or pearly appearance
- Visible branching blood vessels
- Non-healing sores or ulcers

“

“AI aids in early and accurate diagnosis by detecting skin structures and abnormalities invisible to the naked eye.”

—

DERMATOSCOPY CONCLUSION

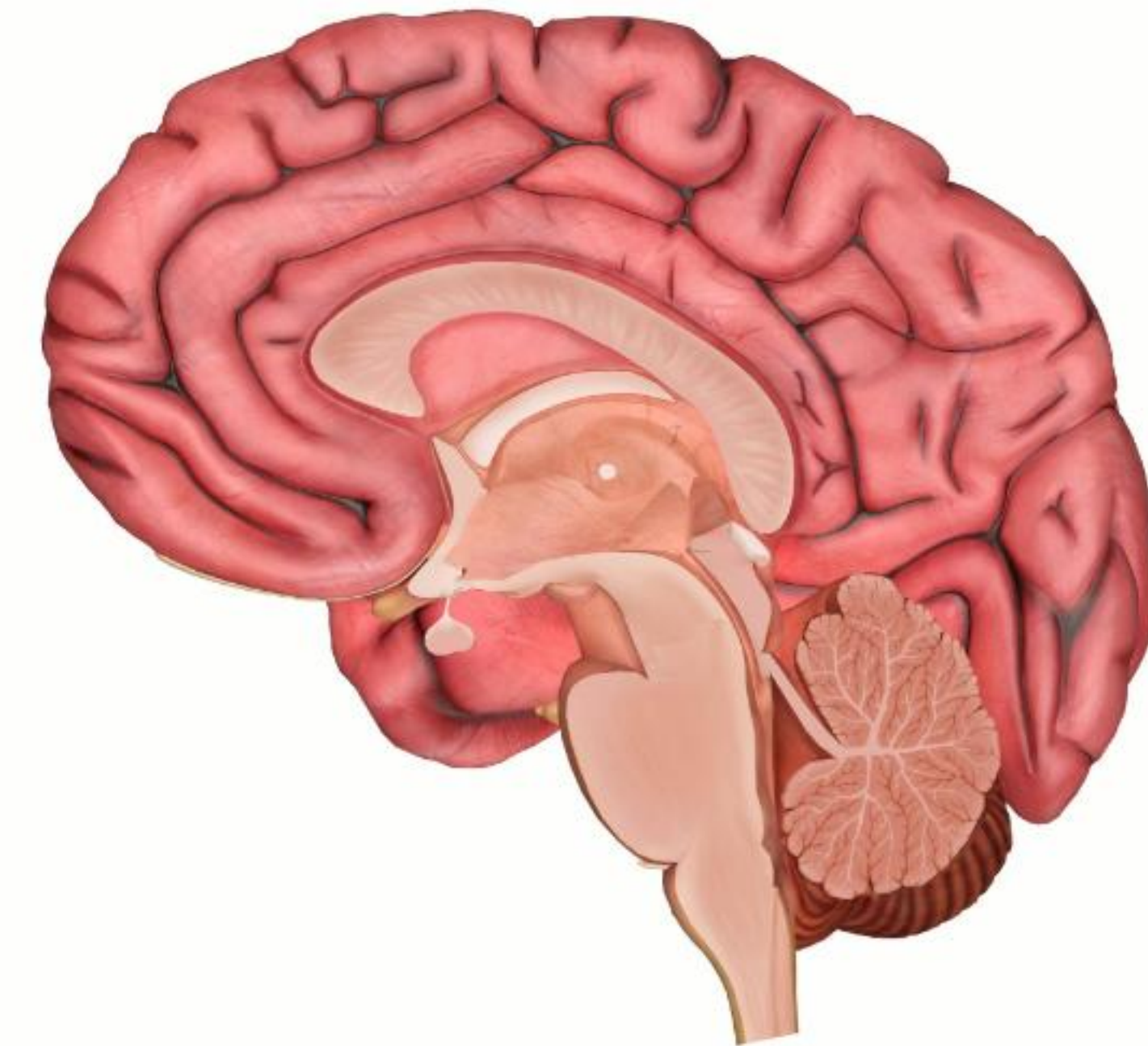
SECTION 02

MRI SCANNING

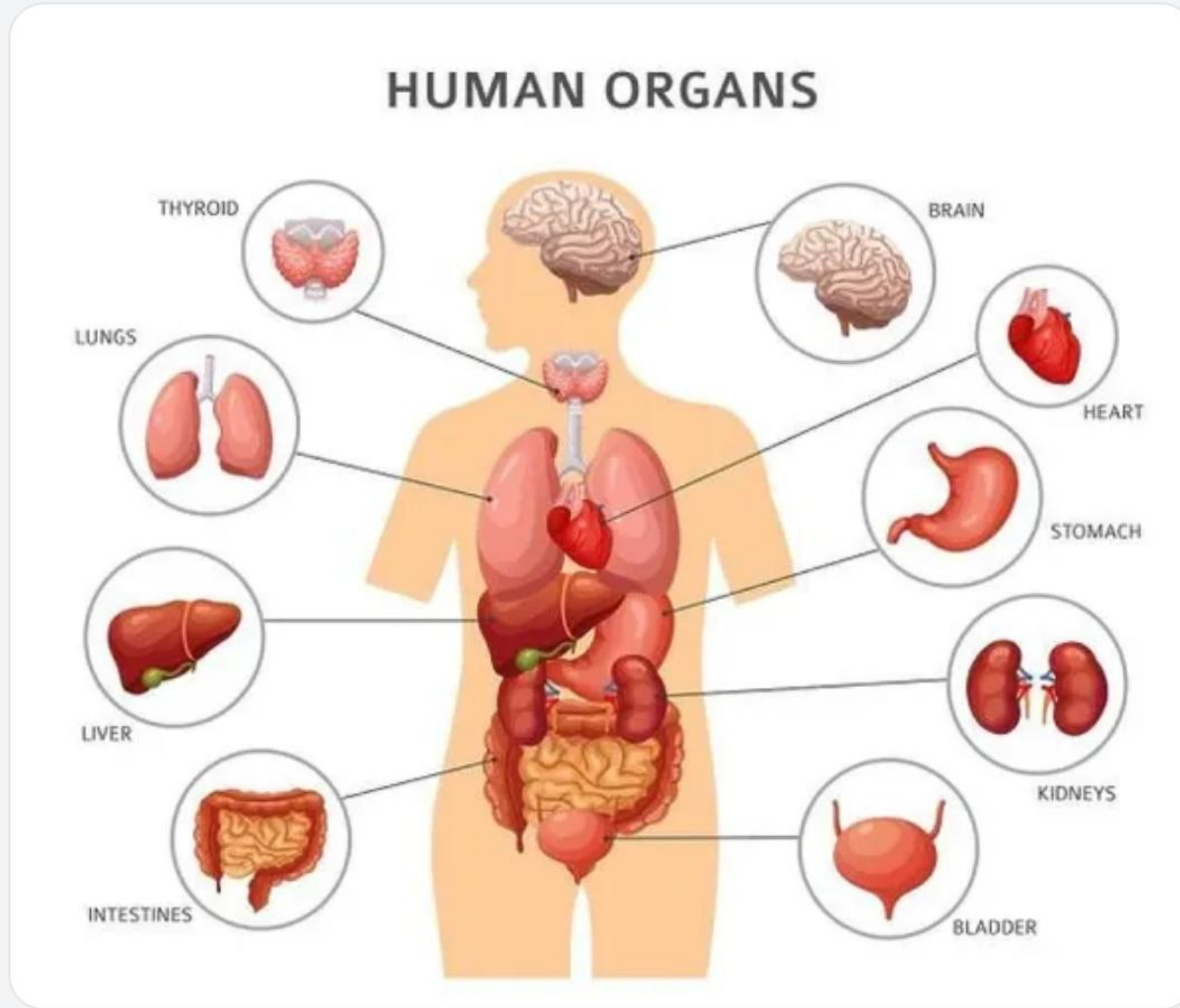
MAGNETIC RESONANCE IMAGING

Non-Invasive Safety

- Uses magnets and radio waves (No X-Rays)
- Safe and painless internal visualization
- Ideal for soft tissue, spine, and organs
- Supports precision diagnostic planning



INTERNAL ORGANS



Precision Mapping

MRI provides detailed views of critical systems:

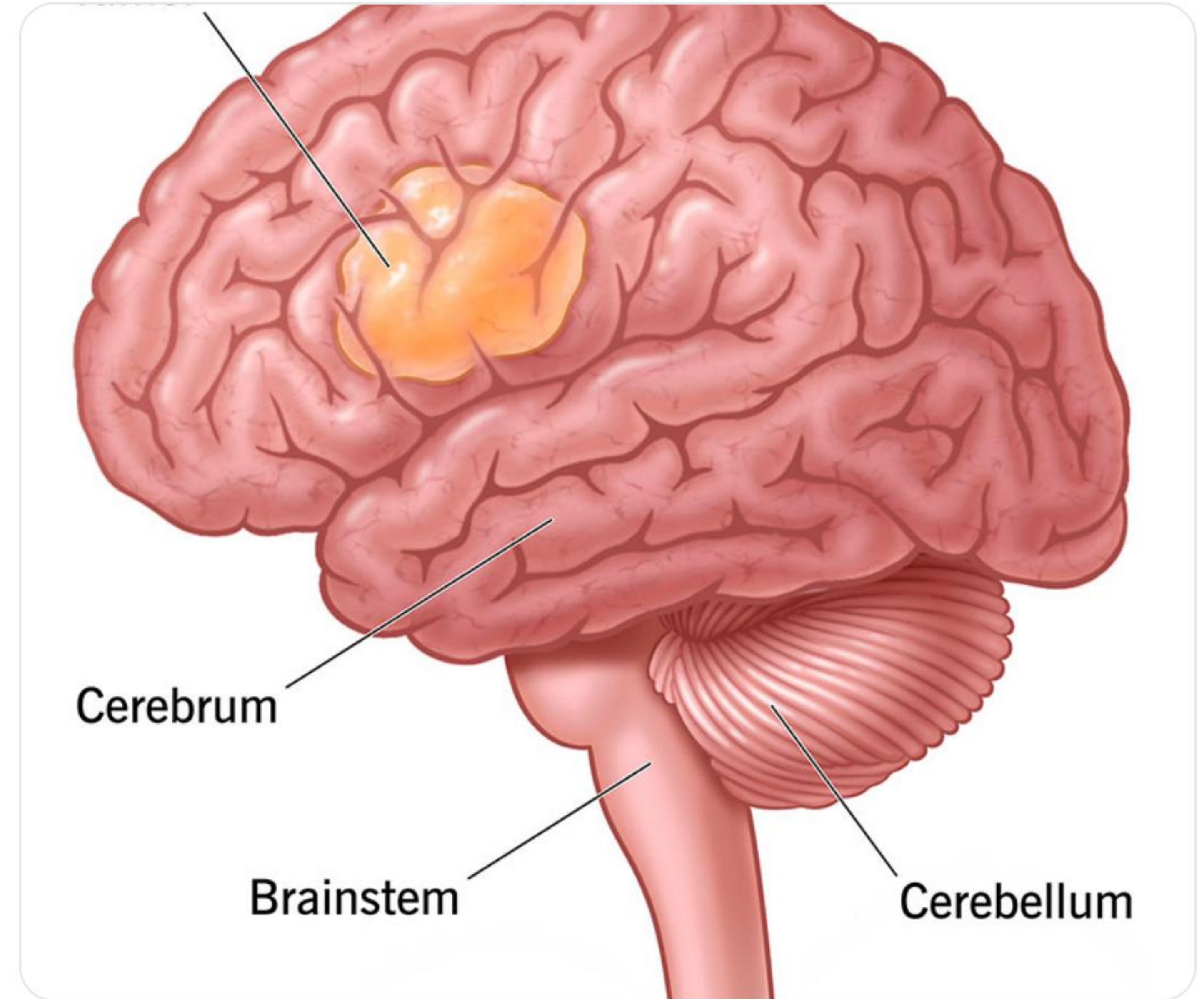
- Brain & Spine
- Liver & Pancreas
- Bladder & Uterus
- Heart & Lungs
- Kidneys & Intestines

BRAIN TUMOR DETECTION

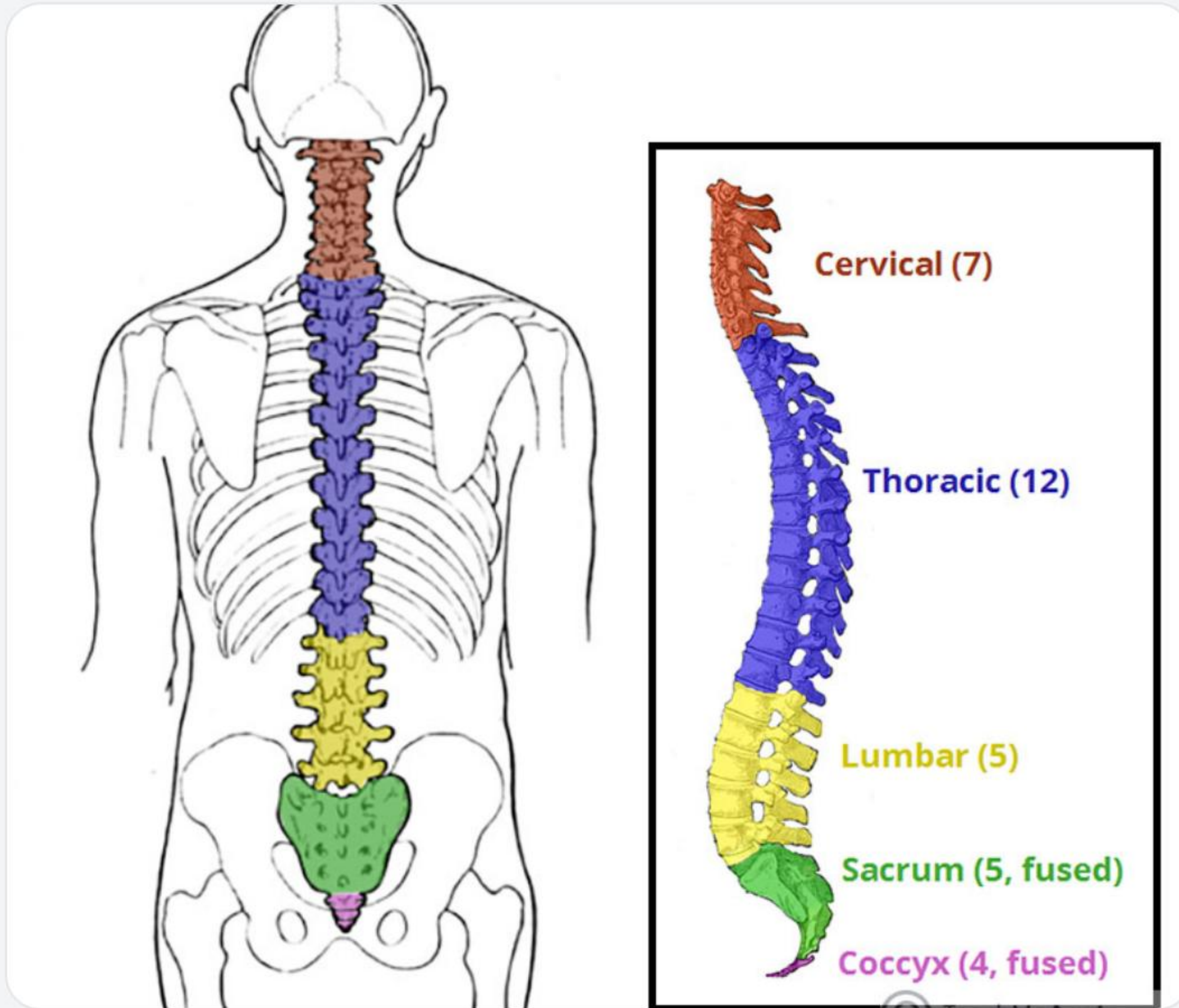
Mass Localization

MRI can detect abnormal growths in the brain, identifying size and effect on surrounding tissue.

- Identifies benign vs malignant masses
- Locates tumor within Cerebrum/Cerebellum
- Essential for surgical planning



SPINAL CORD PROFILING



- **Disc Health:** Detects herniated or bulging discs.
- **Nerve Compression:** Shows pressure on the spinal cord.
- **Trauma:** Identifies ligament injuries or soft tissue damage.
- **Inflammation:** Locates cysts or cysts around the spine.

| HEART & TUMOR DYNAMICS

Cardiac Structure

MRI evaluates the heart's chambers and blood flow, identifying dangerous abnormal growths.



3D Assessment of Muscle Function

 Heart tumor diagram

LIVER, PANCREAS, KIDNEY

 Organs diagram

Abdominal Pathology

Detecting tumors, cysts, or structural abnormalities in the filtration systems.

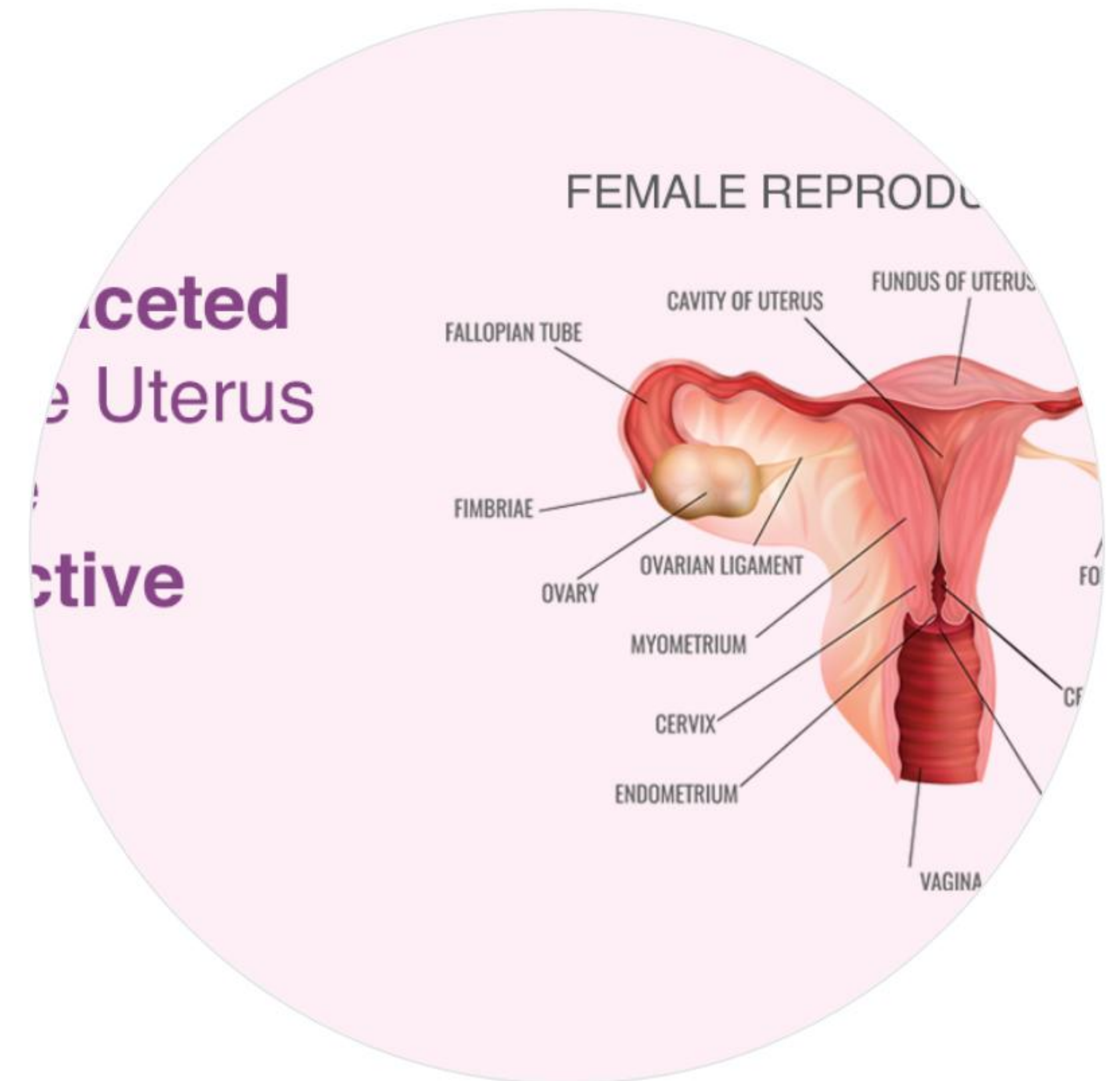
- Shows organ structure and tissue details
- Monitors treatment progress effectively
- Highly sensitive for early detection

REPRODUCTIVE ORGANS

Uterus & Soft Tissue

Detection of cysts, structural abnormalities, or tumors in reproductive systems.

- Detects abnormal growths
- Provides detailed organ structure
- Helps doctors monitor recovery



MRI CONCLUSION

"MRI provides detailed images of soft tissues and internal organs. AI aids faster and accurate detection, helping doctors plan treatment."

SECTION 03

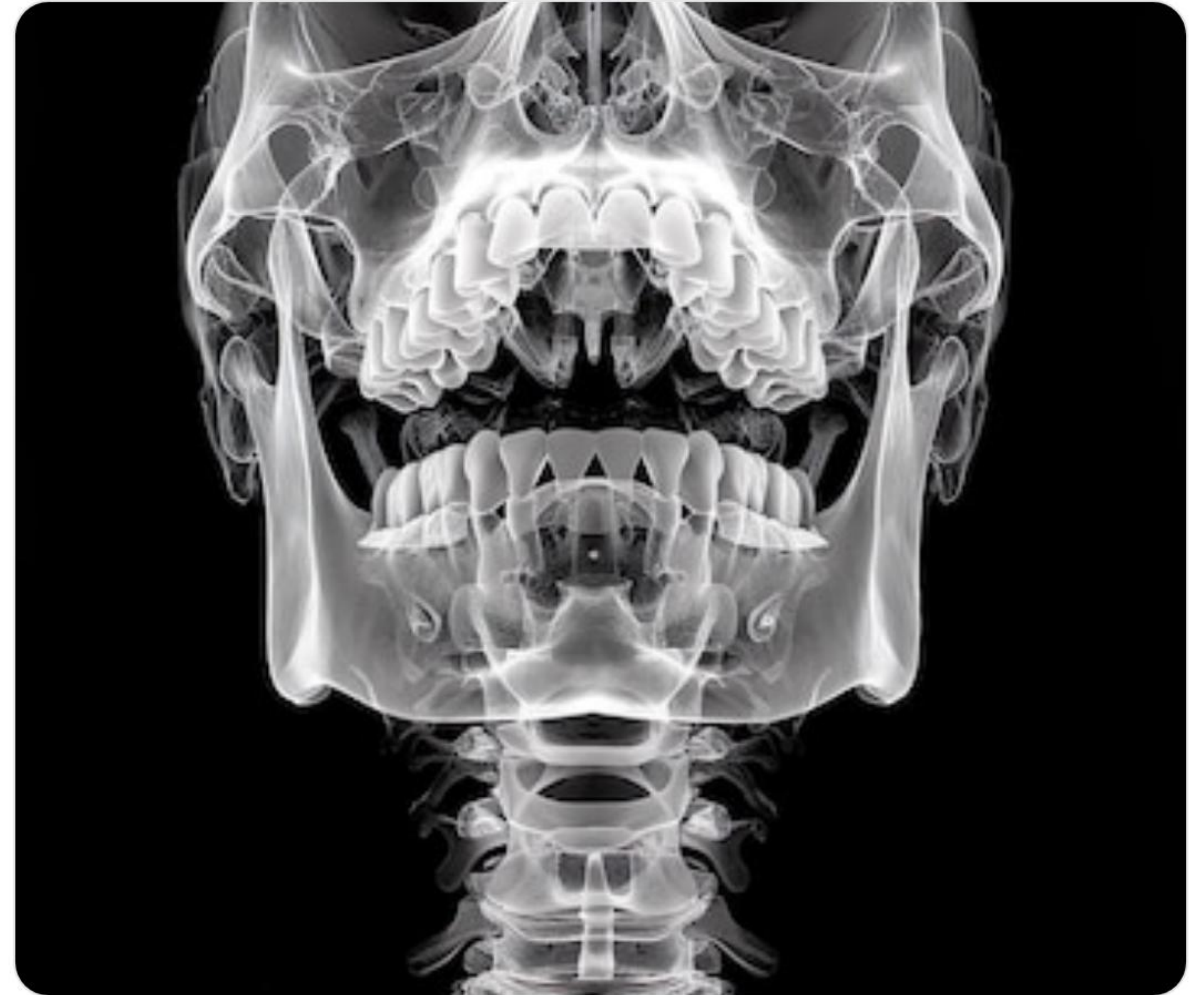
OPG X-RAY

ORTHOPANTOMOGRAM X-RAY

Panoramic Vision

OPG provides a wide, 2D view of the mouth and jaw structure.

- Examines teeth and jaw clearly
- Full upper and lower jaw view
- Quick and simple imaging process
- Essential for initial diagnosis



NORMAL X-RAY



Healthy Alignment

A normal scan shows healthy teeth and proper alignment without signs of infection.

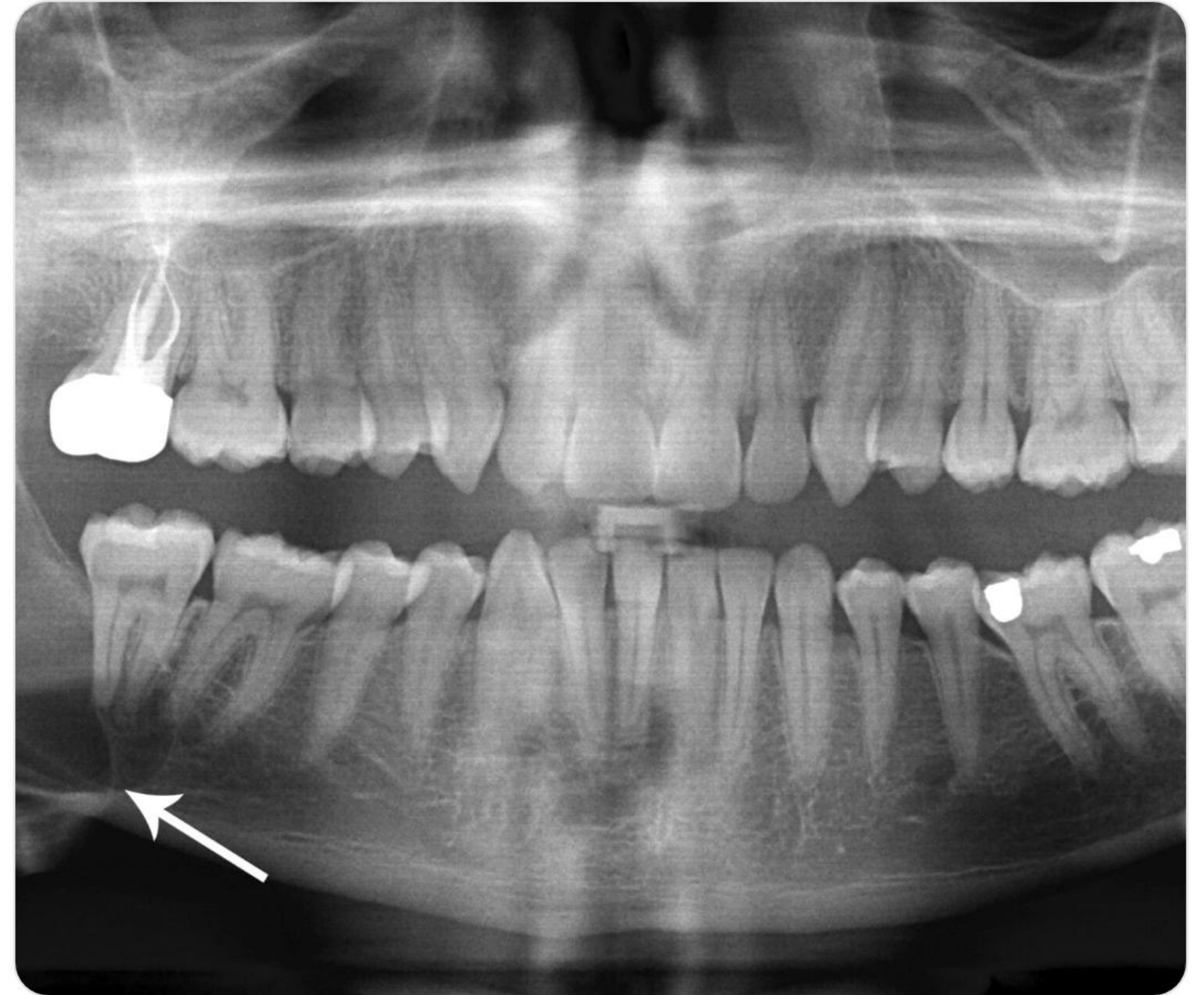
- Properly aligned teeth
- No cavities or infections visible
- Smooth and even jaw bone

PATHOLOGY: CYSTS

Fluid-Filled Sacs

A cyst can grow slowly and damage the surrounding jaw bone.

- Appears as a dark round area
- Can weaken the jaw bone structure
- Often requires surgical removal



PATHOLOGY: JAW FRACTURE

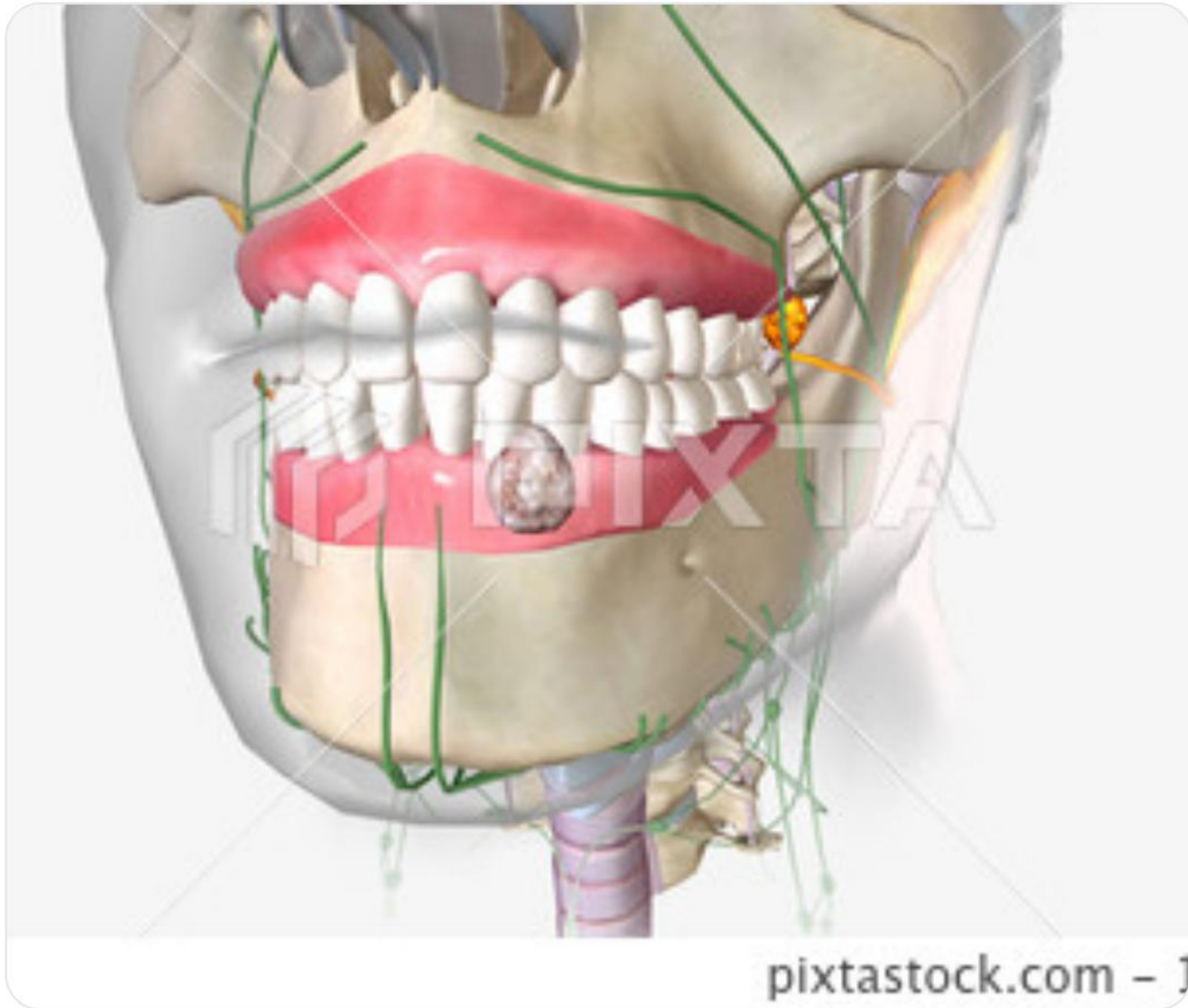
Structural Breaks

Caused by injury or accidents, fractures disrupt the bone line.

- Clear break in the bone line
- Causes pain and swelling
- Affects normal jaw movement



PATHOLOGY: JAW TUMORS



Abnormal Growth

Tumors can grow over time and become serious if not treated.

- Irregular dark or light areas
- Affects surrounding bone and teeth
- Requires immediate evaluation

OPG CONCLUSION

"Panoramic views detect cysts, fractures, and tumors. AI aids in faster and more accurate diagnosis."

| THE POWER OF AI

2030

Vision for Healthcare

Our platform integrates high-resolution imaging with predictive models to achieve 99% diagnostic accuracy in early triage.



Faster Triage

Reduction in diagnostic latency by up to 60% compared to manual review.

QUESTIONS?

Explore the full platform live at the Science Fair.

medical-prediction-94d.caffeine.xyz

THANK YOU

THE DIAGNOSTICS CREW

Deekshith • Advith • Manesh • Sana Akshaya • Aleekhya

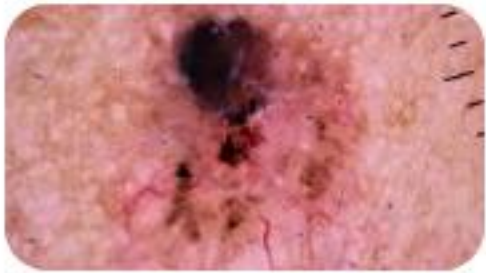


IMAGE SOURCES



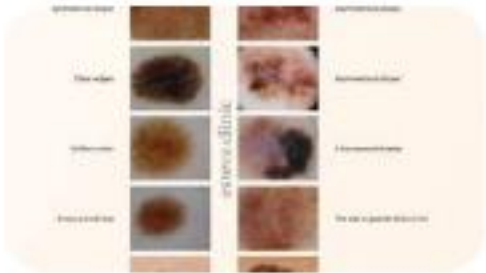
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Source: www.mdpi.com



<https://dermoscopy.wordpress.com/wp-content/uploads/2022/12/boxing-day-case-2.jpg>

Source: dermoscopy.wordpress.com



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Source: www.esteva.com.ua



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Source: www.cancer.org



<https://innerbody.imgix.net/brain.png>

Source: www.innerbody.com



<https://prints.alamy.com/p/780/human-body-internal-organs-stomach-lungs-39554330.jpg.webp>

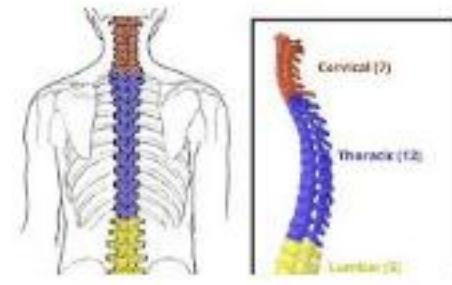
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IMAGE SOURCES



<https://my.clevelandclinic.org/-/scassets/images/org/health/articles/6149-brain-tumor>

Source: my.clevelandclinic.org



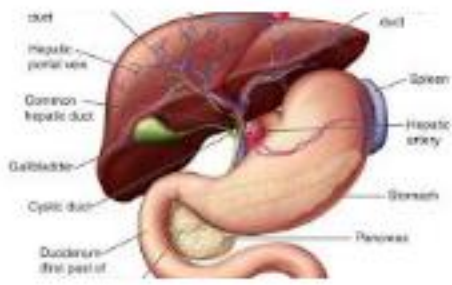
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Source: longhornbrainandspine.com



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Source: www.youtube.com



https://www.hopkinsmedicine.org/-/media/images/health/1_conditions/liver-gallbladder-and-pancreas/liver-anatomy.png

Source: www.hopkinsmedicine.org



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Source: oasisindia.in



https://img.freepik.com/premium-photo/xray-human-skull-with-focus-teeth-jaw-structure_1046712-4118.jpg?w=360

Source: www.freepik.com

IMAGE SOURCES



<https://healthysmiledental.com.au/wp-content/uploads/2019/08/OPG-machine-Healthy-Smile-dental-Underewood.jpg>

Source: healthysmiledental.com.au



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